

Denumerable Markov chains

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Duration: 3 hours

1 Questions on Markov chains

Let $(X_n)_{n \geq 0}$ be a 2 states Markov chain, on $X = \{1, 2\}$. The most general stochastic matrix P of $(X_n)_{n \geq 0}$ can be written as:

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

with $\alpha, \beta \in [0, 1]$.

1. Compute the invariant probability distribution(s). Give a necessary and sufficient condition for uniqueness of the invariant probability distribution.
2. Assume that $\alpha + \beta > 0$. Prove that, for all $n \geq 0$

$$P^n = \begin{pmatrix} 1 - p & p \\ 1 - p & p \end{pmatrix} + \rho^n \begin{pmatrix} p & -p \\ -(1 - p) & 1 - p \end{pmatrix}$$

for some p and ρ which shall be computed.

3. Assume that $\alpha + \beta \in]0, 2[$. Prove that for any initial probability measure μ on X , the limit $\lim_{n \rightarrow +\infty} \mu P^n$ exists and does not depend on μ .
4. When is (X, P) irreducible? When is (X, P) periodic? When does P have only one closed essential class? When does P have one transient class?

2 Questions on Galton-Watson process

We denote $\mathbb{N} := \{0, 1, 2, \dots\}$ and fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $(N_{k,n})_{k,n \geq 1}$ be a family of $\mathbb{N}$ -valued, independent and identically distributed random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$.

We consider the sequence $(Z_n)_{n \geq 0}$ of random variables defined by $Z_0 = 1$ and, for $n \geq 1$,

$$Z_n = \begin{cases} N_{1,n} + N_{2,n} + \dots + N_{Z_{n-1},n} & \text{if } Z_{n-1} \neq 0, \\ 0 & \text{if } Z_{n-1} = 0. \end{cases}$$

Assume that the *offspring distribution* μ , that is the distribution of the variables $N_{k,n}$, is *non-degenerated*, i.e. $\mu\{k\} \neq 1$ for any $k \geq 0$; in the sequel, f denotes the generating function of μ .

The sequence $(Z_n)_n$ is the *Galton-Watson process* with offspring distribution μ ; the variable Z_n represents the size of the n -th generation of some population.

1. Give the definition of f according to μ . Why is f defined (at least) on $[0, 1]$?

We set $m := \sum_{k \geq 0} k\mu\{k\}$. What is the relation between m and f ? (a detailed proof is required)

2. What is the relation between the generating function f_n of Z_n and the function f ? (a detailed proof is required)
3. Let e_n be the extinction probability at the n^{th} generation: $e_n = \mathbb{P}(Z_n = 0)$.
 - (a) Express e_n with f_n .
 - (b) Prove that the sequence $(e_n)_n$ is non decreasing and that its limit e satisfies $f(e) = e$.
 - (c) Prove that e is the smallest non-negative fixed point of f .

4. What is the condition on m which ensures that $e = 1$?
5. We assume that $\mu\{0\} = 1 - \alpha$ and $\forall k \geq 1 \mu\{k\} = \alpha p(1 - p)^{k-1}$, where $0 < \alpha < 1$ and $0 < p < 1$ are two fixed parameters.
 - (a) Check that the measure μ is a probability measure on $\mathbb{N}$.
 - (b) Compute the generating function f of μ (in terms of the parameters α and p).
 - (c) Express the expectation of μ (in terms of the parameters α and p).
 - (d) Check that the Galton-Watson process gets extinct almost surely if and only if $\alpha \leq p$.

3 Exercise

We consider the Markov chain $(X_n)_{n \geq 0}$ on $\{1, 2, 3, 4, 5, 6, 7\}$ with transition matrix

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2/3 & 1/3 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 & 0 \\ 1/4 & 0 & 0 & 0 & 0 & 1/2 & 1/4 \\ 0 & 1/4 & 0 & 0 & 0 & 1/4 & 1/2 \end{pmatrix}.$$

1. Represent the graph of $(X_n)_{n \geq 0}$.
2. What are the irreducible classes of this Markov chain?
 - (a) What are the essential classes and the transient ones?
 - (b) Are these classes aperiodic or not?
3. Which states are recurrent and which ones are transient?
4. Compute the fundamental matrix N of this Markov chain.
5. We assume now that $X_0 = 7$. What is the probability that the chain reaches the state $\{2\}$ when it leaves the set $\{6, 7\}$?
6. We assume now that $X_0 \in \{4, 5\}$. What is the probability that the chain reaches the state $\{3\}$ when it leaves the set $\{4, 5\}$?

4 Exercise

We consider an urn and the following mechanism. At step $n \geq 0$, with probability $p > 0$, a ball is added to the urn; with probability $q = 1 - p > 0$, a ball is removed from the urn when the urn is not empty and nothing happens when the urn is empty. For any $n \geq 0$, let X_n be the number of balls in the urn at step n .

1. Explain why $(X_n)_{n \geq 0}$ is a Markov chain. Compute its transition probability matrix P . Is this chain irreducible? aperiodic?
2. Prove that $(X_n)_{n \geq 0}$ has a unique stationary measure $\nu = (\nu_k)_{k \geq 0}$ on $\mathbb{N}$ such that $\nu_0 > 0$ (then $\nu_k > 0$ for any $k \geq 0$).
3. Prove that P is reversible with respect to ν , which means that $\nu(k)P(k, \ell) = \nu(\ell)P(\ell, k)$ for any $k, \ell \geq 0$.
4. We assume $p < 1/2$.
 - (a) Prove that ν is finite.
 - (b) Is the chain recurrent in this case? Why?
 - (c) Compute the asymptotic proportion of time the urn is empty.
 - (d) Compute the mean time to empty again the urn starting from an empty urn.